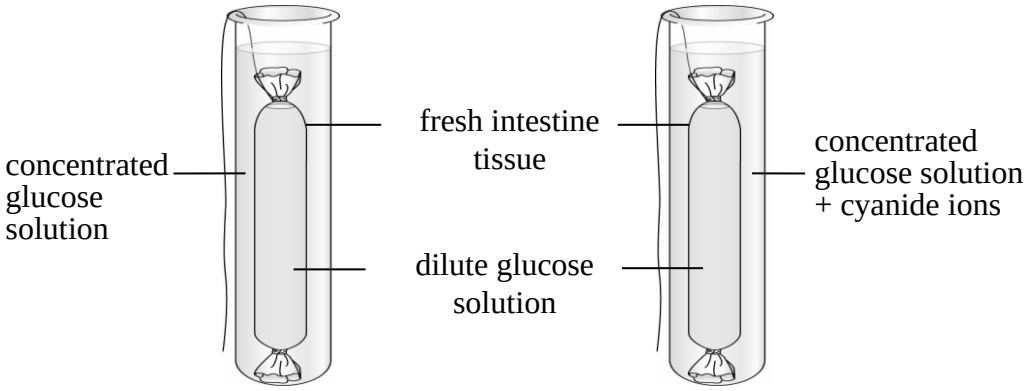
Multiple-choice questions

|!|BCEMA0040000003|!| ★★ SI

A student carried out an experiment using sections of intestine freshly isolated from a rat. Cyanide ions inhibit enzymes involved in respiration. The diagram below shows a set-up.



After few hours, only the concentration of glucose solution inside the intestine in test tube A increased. What can be deduced from the result?

A. Oxygen is necessary for the movement of glucose across intestinal wall.
B. Cyanide ions inhibit the respiratory enzymes by altering the shapes of their active sites.
C. Glucose molecules enter the small intestine by active transport.
D. There is net movement of water molecules from the concentrated glucose solution into the intestine of test tube A.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000004|!| ★

Lipids are

A. components of plant cell walls.
B. involved in active transport.
C. transported by rough endoplasmic reticulum.
D. made up of hydrogen atoms and oxygen atoms in the ratio greater than 2 : 1.

##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000005|!| ★

Which of the following processes is possible due to the fluidity of the cell membrane?

A. osmosis
B. active transport
C. phagocytosis
D. diffusion

□

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0040000006|!| ★

An animal cell placed in liquid X swells up and bursts. What will happen to a plant cell placed in the same liquid?

A. It will become turgid.
B. It will become flaccid.
C. It will become plasmolysed.
D. It will not be affected.

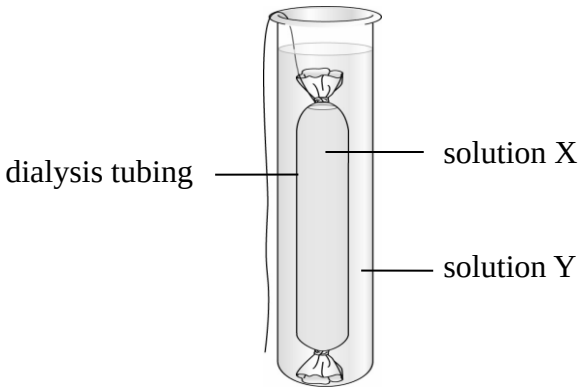
□

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0040000007|!| ★★

An experiment was carried out to investigate osmosis using the following set-up.



The initial mass and final mass of the tubing was measured as shown in the table below.

Initial mass	12.0 g
Final mass	9.5 g

© 2019 Aristo Educational Press Ltd.

These results would be obtained when

A. X is a 5% sucrose solution and Y is distilled water.

B. X is a 5% sucrose solution and Y is a 10% sucrose solution.

C. X is a 10% sucrose solution and Y is a 5% sucrose solution.

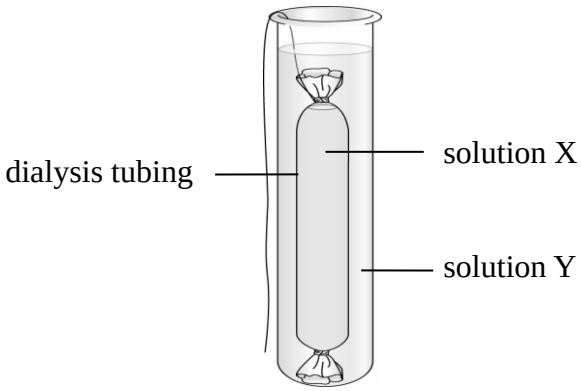
D. X is a 10% sucrose solution and Y is distilled water.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0040000008|!| ★★ SI

The diagram below shows an apparatus used to investigate osmosis.



Which of the following combinations shows the solutions used in the investigation and the correct results?

	Solution X	***Solution Y***	***Change of volume in dialysis tubing***
A.	10% sucrose	5% sucrose	decrease
B.	distilled water	5% sucrose	increase
C.	5% sucrose	10% sucrose	increase
D.	15% sucrose	10% sucrose	increase

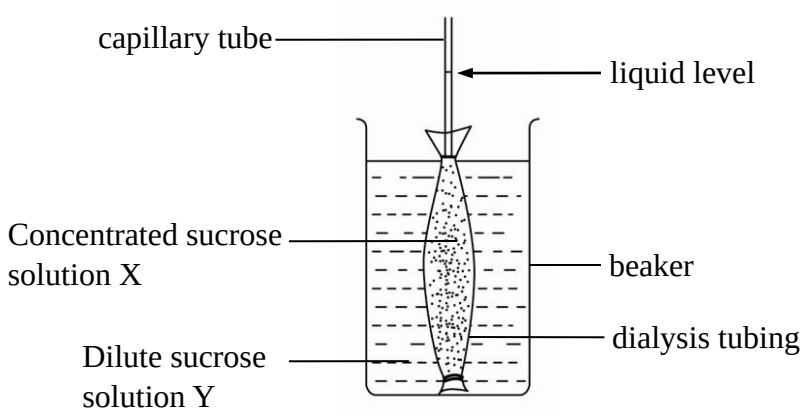
##D##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000010|!| ★

The following set-up was used to investigate osmosis.



30 minutes after the beginning of the experiment, the liquid level in the glass tube had risen because

A. water molecules pass from X to Y.
B. water molecules pass from Y to X.
C. more water molecules pass from X to Y than from Y to X.
D. more water molecules pass from Y to X than from X to Y.

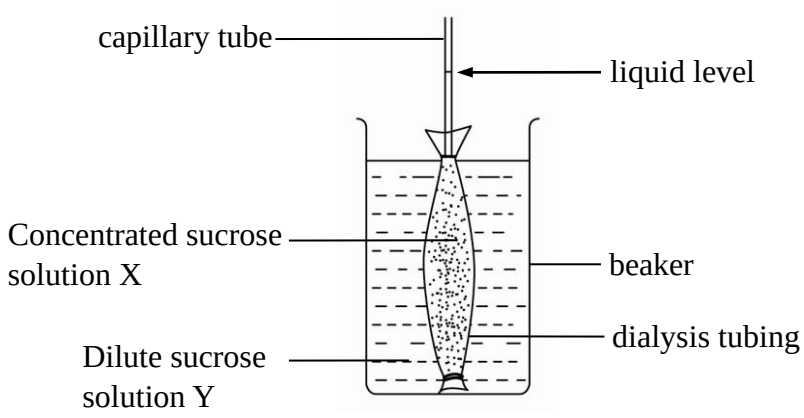
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0040000011|!| ★★ **SI**

The following set-up was used to investigate osmosis.



30 minutes after the beginning of the experiment, the liquid level in the glass tube had risen. The liquid level would rise at a faster rate if

(1) a longer glass tube was used.
(2) a narrower glass tube was used.

© 2019 Aristo Educational Press Ltd.

(3) the sucrose solution X was made more concentrated.
(4) the sucrose solution Y was made more concentrated.

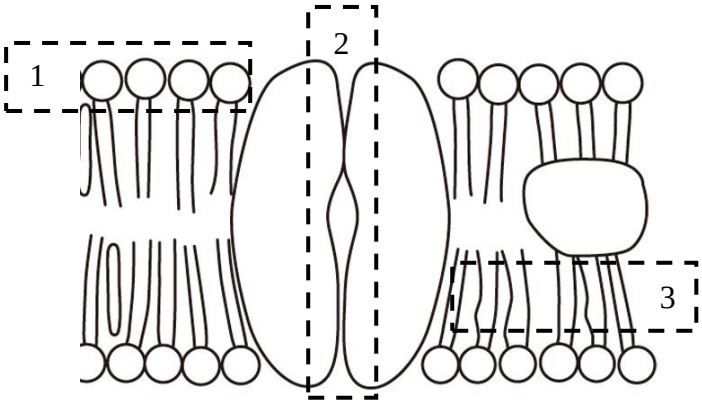
A. (1) and (3) only
B. (2) and (3) only
C. (2) and (4) only
D. (1), (2) and (4) only

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0040000012|!| ★

The diagram below shows a small part of the cell membrane.



Which region(s) is/are hydrophobic?

A. 2 only
B. 3 only
C. 1 and 3 only
D. 2 and 3 only

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0040000013|!|

Which of the following components of the cell membrane allows the movement of ions across it?

A. fatty acid
B. protein
C. phospholipid
D. cholesterol

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0040000014|!|

Which of the following combinations correctly shows the function of the molecule of the cell membrane?

	Transporting ions	*Stabilizing the hydrophobic layer*	*Acting as recognition marker*
A.	protein	cholesterol	glycoprotein
B.	glycoprotein	phospholipid	protein
C.	phospholipid	protein	cholesterol
D.	cholesterol	glycoprotein	phospholipid

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0040000015|!|

Which of the following correctly matches the cell membrane component with its function?

	Cell membrane component	*Function*
A.	phospholipid bilayer	allows water-soluble molecules to pass through
B.	glycoprotein	involves in the active transport of molecules
C.	carrier protein	for cell recognition
D.	cholesterol	stabilizes the membrane structure

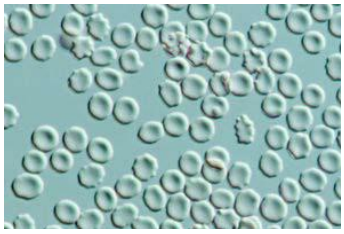
##D##

Skill assessed	Mark
Understanding basic concepts	1

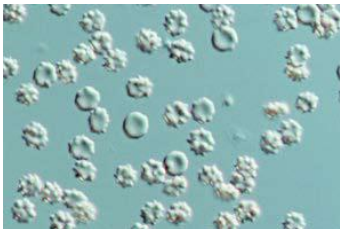
© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000016|!| ★

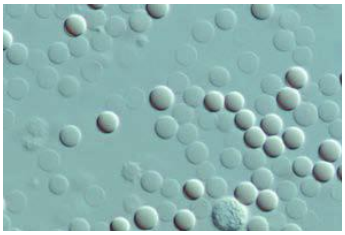
The following photomicrographs show the appearance of red blood cells immersed in different concentrations of sucrose solution (A, B and C).



sucrose solution A



sucrose solution B



sucrose solution C

Which of the following correctly arranges the concentrations of sucrose solutions A, B and C in descending order?

A. A > C > B
B. B > A > C
C. C > A > B
D. C > B > A

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000018|!|

What will happen if an onion epidermal cell is placed in distilled water?

A. It will burst.
B. It will become turgid.
C. It will shrink.
D. It will remain normal in shape.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0040000020|!|

Which of the following processes involve(s) active transport?

(1) the absorption of glucose by cells lining the small intestine in humans
(2) the absorption of carbon dioxide by leaf cells
(3) the uptake of mineral ions into plant root cells from the soil

© 2019 Aristo Educational Press Ltd.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0040000021|!| ★

Which of the following cells take(s) in particles by phagocytosis?

(1) red blood cell
(2) root cell
(3) *Amoeba* cell

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

☐

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0040000022|!|

What happens when a plant cell is placed in a hypertonic solution?

A. The cell becomes plasmolysed.
B. The cell bursts.
C. The cell becomes turgid.
D. The cell remains unchanged in size.

☐

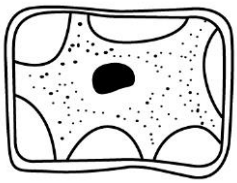
##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000023|!| ★

The diagram below shows a plant cell immersed in a 15% sucrose solution.



What property of the sucrose solution causes the appearance of the plant cell?

A. It has a lower water potential than that of the cytoplasm of the cell.
B. It has a higher water potential than that of the cytoplasm of the cell.
C. It pushes the cell membrane inwards from the cell wall.
D. It causes the breakdown of the cell membrane.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000024|!|

Which of the following processes involve(s) carrier proteins?

(1) phagocytosis
(2) osmosis
(3) active transport

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0040000025|!|

The rate of diffusion can be increased by

(1) increasing the temperature.
(2) increasing the surface area over which diffusion occurs.
(3) decreasing the concentration gradient between the two regions.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0040000026|!|

Which of the following statements about active transport is correct?

(1) Active transport relies on carrier proteins present in the cell membrane.
(2) Active transport can occurs against a concentration gradient.
(3) Active transport can only occur in the presence of oxygen.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000028|!| ★

Which of the following is/are differentially permeable?

(1) potato skin
(2) dialysis tubing
(3) chicken crop

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

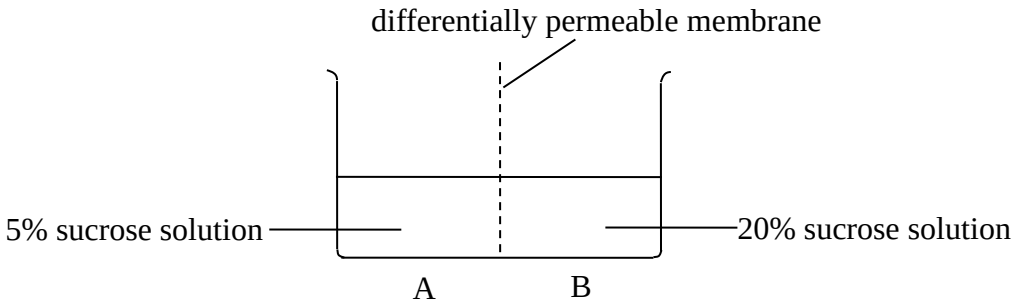
□

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0040000029|!| ★

The following diagram shows an experimental set-up used to study osmosis.



Which of the following statements about the set-up is/are correct?

(1) Water molecules only move from A to B.
(2) Sucrose molecules move from B to A.
(3) The liquid level in B will rise.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

□

##B##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000030|!| ★

A dialysis tubing containing a mixture of glucose and sodium chloride solutions is immersed in a beaker of distilled water. An hour later, both glucose and sodium chloride are detected outside the tubing. Which of the following processes is/are involved in the movement of glucose and sodium chloride across the tubing?

(1) diffusion
(2) osmosis
(3) active transport

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

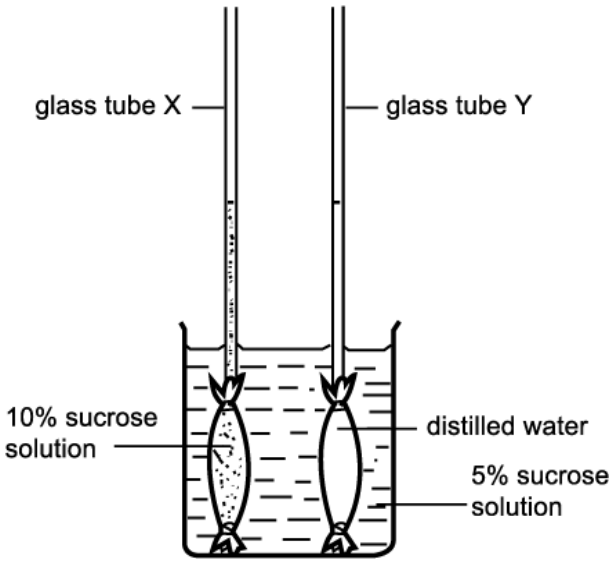
□

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0040000034|!| ★

The diagram below shows an apparatus in which the dialysis tubing is made of a differentially permeable membrane.



What will happen to the liquid levels in the glass tubes after one hour?

	Glass tube X	*Glass tube Y*
A.	falls	rises
B.	remains unchanged	rises
C.	rises	falls
D.	rises	remains unchanged

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000036|!| **Cross topic**

Which of the following processes involves active transport?

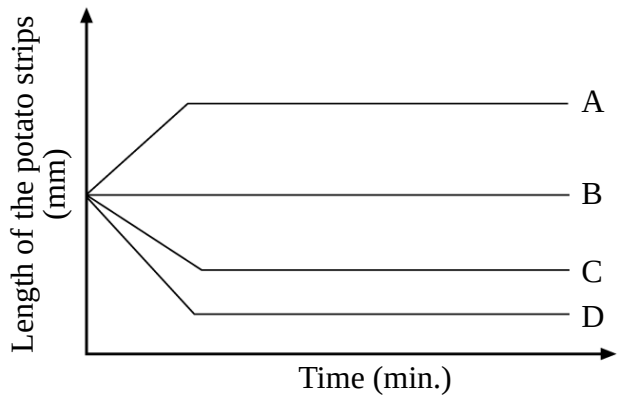
A. uptake of water by root cells
B. absorption of digested food in the small intestine
C. exchange of gases in air sacs
D. ingestion of bacteria by white blood cells

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0040000037|!| ★

In an experiment, four potato strips of the same length and same diameter were cut from the same potato and immersed in sucrose solutions of different concentrations. At regular intervals, the length of each potato strip was measured. The results are shown in the graph below.



Which curve shows the result of a potato strip in the sucrose solution of the highest water potential?

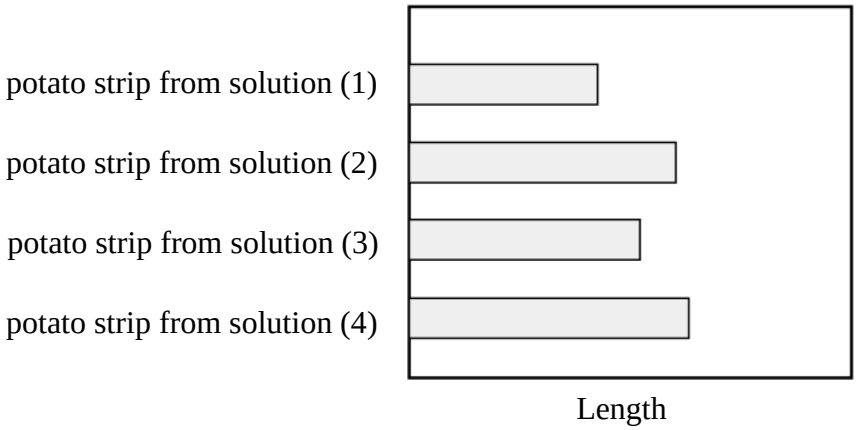
© 2019 Aristo Educational Press Ltd.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000038|!| ★

Four strips of potato were cut to the same length and thickness. They were immersed in four solutions of different water potentials respectively. After an hour, the lengths of the potato strips were measured. The results are shown below.



Which of the following conclusions can be drawn from the results?

A. Solution (1) has the highest water potential.
B. Solution (2) has a higher water potential than solution (4).
C. Solution (3) has a lower water potential than solution (4).
D. Solution (4) is distilled water.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000039|!|

Which of the following statements correctly describe the features of active transport?

(1) It requires ATP.
(2) It requires the help of carrier proteins.
(3) It can move particles along a concentration gradient.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only

© 2019 Aristo Educational Press Ltd.

D. (1), (2) and (3)

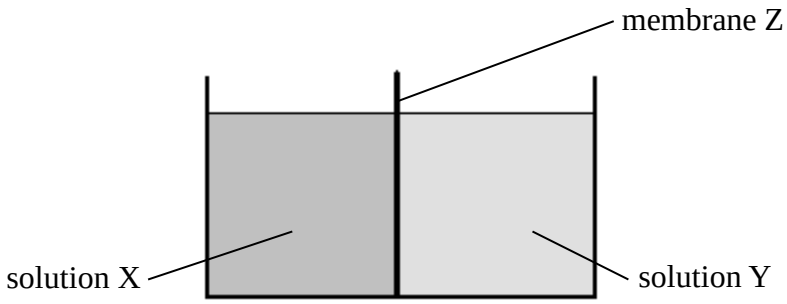
☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0040000041|!| ★

The diagram below shows an experimental set-up, with solutions X and Y being separated by membrane Z.



Which of the following combinations would lead to a rise in the liquid level in solution Y?

	Solution X	***Solution Y***	***Substances that make up membrane Z***
A.	5% sucrose solution	10% sucrose solution	plastic wrap
B.	10% sucrose solution	5% sucrose solution	chicken crop
C.	distilled water	10% sucrose solution	fish swim bladder
D.	10% sucrose solution	distilled water	dialysis membrane

☐

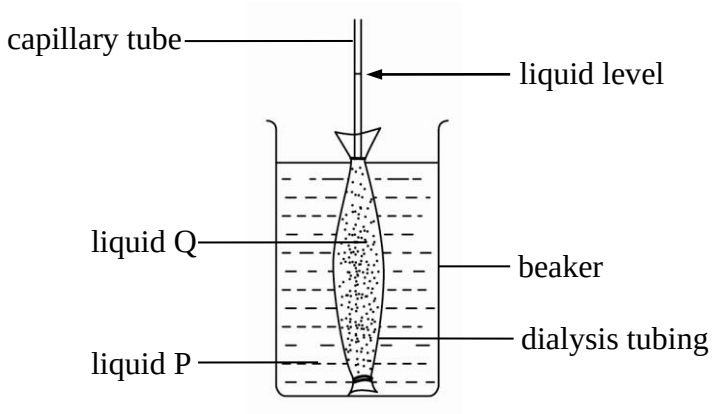
##C##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000044|!| ★

The diagram below shows an experimental set-up for the study of osmosis.



Which of the following combinations of liquids P and Q will cause the liquid level of the capillary tube to drop in the greatest extent after 15 minutes?

	Liquid P	***Liquid Q***
A.	20% sucrose solution	distilled water
B.	distilled water	10% glucose solution
C.	5% sucrose solution	10% sucrose solution
D.	15% sucrose solution	10% sucrose solution

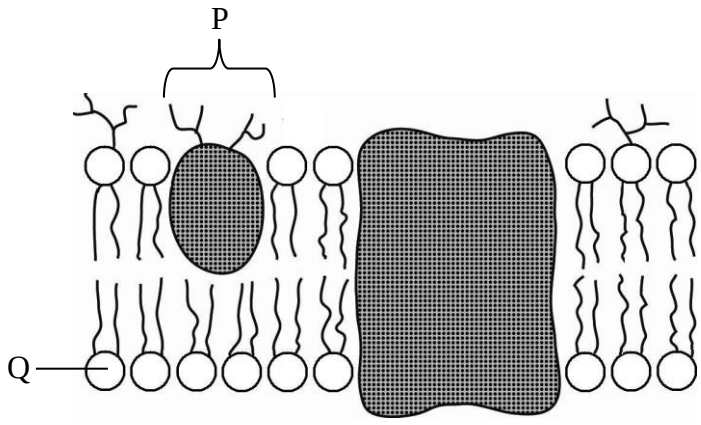
□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000045|!|

The diagram below shows the cross sectional view of a plasma membrane.



Which of the following combinations correctly identifies structures P and Q?

	Structure P	***Structure Q***
A.	glycoprotein	phosphate
B.	glycoprotein	fatty acid
C.	cholesterol	phosphate
D.	cholesterol	fatty acid

☐

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000046|!|

Which of the following chemical substances is ***not*** a component of the cell membrane?

A. amino acid
B. fatty acid
C. phosphate
D. glycogen

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0040000047|!|

Which of the following statements about the cell membrane is/are correct?

(1) It is freely permeable to water and gases.
(2) It is mainly made up of phospholipids.
(3) It can allow amino acid to pass through freely.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0040000050|!|

Haemolysis occurs when red blood cells are placed in

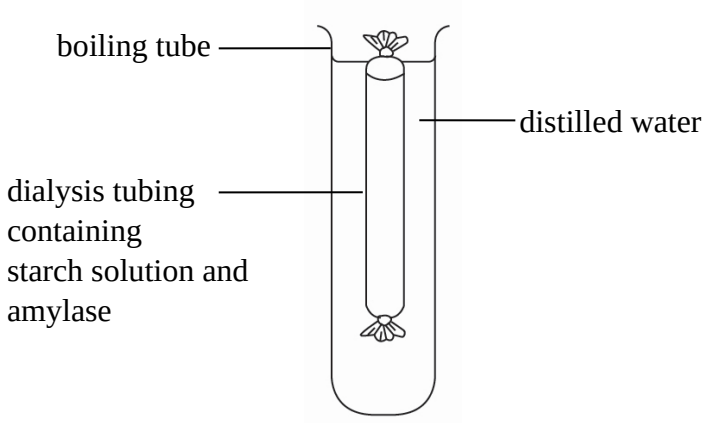
A. a solution that is isotonic to the blood.
B. a solution that is isotonic to the cytoplasm of red blood cells.
C. distilled water.
D. concentrated sucrose solution.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0040000051|!| ★★ SI

Directions: The following two questions refer to the set-up below, which is used to demonstrate osmosis.



The set-up was left undisturbed for one hour. Afterwards, the water in the boiling tube was tested separately using the iodine test, Benedict's test and a protein test strip. Which of the following correctly shows the results?

	Iodine test	*Benedict's test*	*Protein test strip*
A.	blue-black	brick-red precipitate	positive
B.	brown	brick-red precipitate	negative
C.	blue-black	blue	negative
D.	brown	blue	positive

##B##

Skill assessed	Mark
Applying concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0040000052|!| ★★ SI

Which of the following steps can ensure that the results obtained is valid?

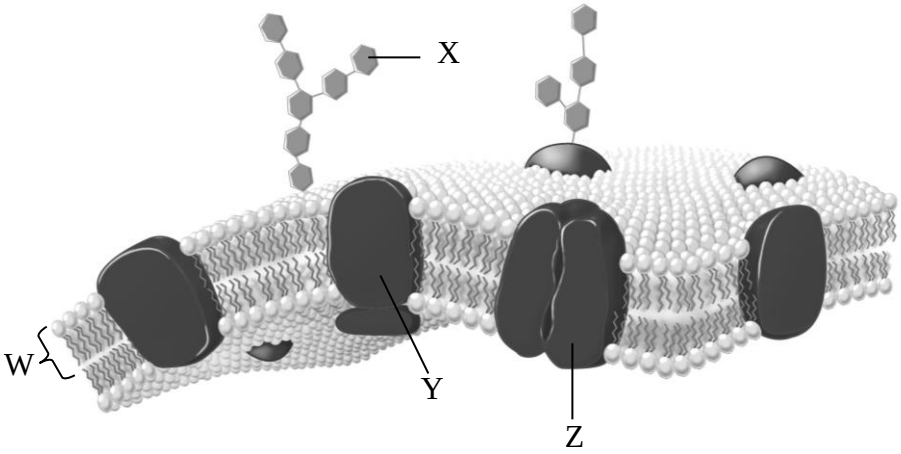
A. Shake the dialysis tubing at regular intervals.
B. Stir the distilled water in the boiling tube at regular intervals.
C. Put the entire set-up in a water bath at 37 ºC.
D. Rinse the outside of the dialysis tubing with distilled water before it is placed into the boiling tube.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0040000053|!| ★

Directions: The following two questions refer to the diagram below, which shows the cell membrane of an epithelial cell in the air sac.



Which part(s) contribute(s) the differential permeability of the cell membrane?

A. W only
B. W and Z only
C. X, Y and Z only
D. W, X and Z only

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0040000054|!| ★

The diffusion of oxygen through the epithelial cell is likely to be achieved via

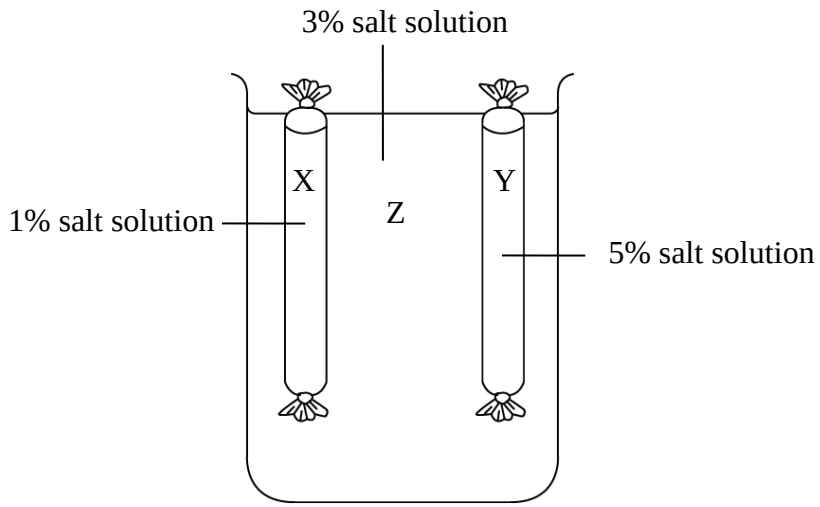
A. W
B. X
C. Y
D. Z

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0040000055|!| ★★ SI

Directions: The following two questions refer to the diagram below, which shows an experiment on osmosis. Two solutions of different salt concentrations (X and Y) are enclosed in two dialysis tubings. Both are immersed in a 3% salt solution (Z).



There is net water movement from

A. Z to both X and Y.
B. X to Z and Z to Y.
C. X and Y to Z.
D. Z to X and Y to Z.

##B##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0040000056|!| ★★ SI

After 24 hours, the concentration of salt solution in Z will be

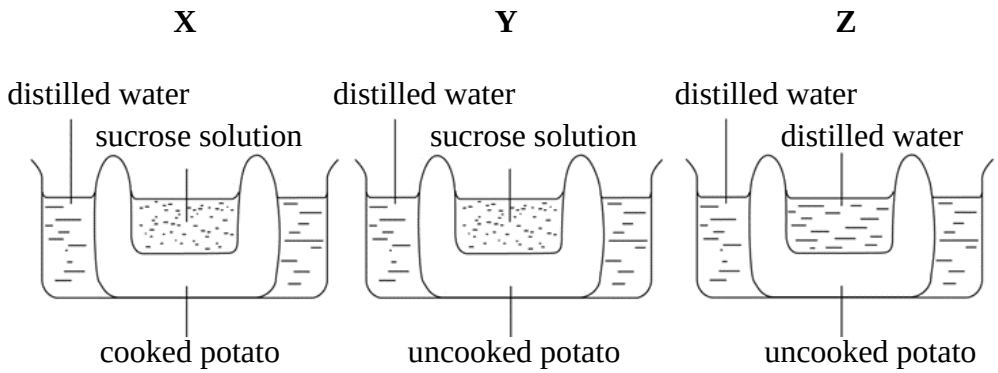
A. greater than X and less than Y.
B. greater than Y and less than X.
C. greater than either X or Y.
D. the same as X and Y.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0040000057|!| ★★ SI

Directions: The following three questions refer to the diagram below which shows an experiment on osmosis. In the experiment, three peeled potatoes X, Y and Z were prepared. X has been cooked in boiling water. A central cavity was made in each potato. Each of the potatoes was then put into a small trough of distilled water. Sucrose solution was poured into the cavities of X and Y, while the cavity of Z was filled with distilled water.



Which of the following correctly shows the permeability of the cell membranes of the cells in potato cups X and Y?

	Potato cup X	***Potato cup Y***
A.	impermeable	differentially permeable
B.	freely permeable	freely permeable
C.	freely permeable	differentially permeable
D.	differentially permeable	impermeable

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0040000058|!| ★★ **SI**

What would be the observation after 12 hours?

A. Only the level of sucrose solution in the cavity of potato Y would have risen.
B. The level of distilled water surrounding potatoes X and Y would have risen.
C. The level of sucrose solution in the cavities of potatoes X and Z would have dropped.
D. The levels of sucrose solution and distilled water in the cavities of potatoes X, Y and Z would remain unchanged.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0040000059|!| ★★ **SI**

The results of this experiment would be as shown in the above question because

(1) potatoes are impermeable so water cannot pass through.
(2) cooking has destroyed the cell membrane structure.
(3) osmosis does not occur when there is no difference in the water potential on the two sides of the living tissue.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000060|!| ★

Directions: The following two questions refer to the experiment below. In the experiment, three potato strips of 5 cm long were placed in three sucrose solutions of different concentrations. Two hours later, the lengths of the three potato strips were measured and recorded in the table below.

Sucrose solution	Final length of the potato strip (cm)
A	5.0
B	4.8
C	5.5

The ascending order of the water potential of these sucrose solutions is

A. A, C, B.
B. B, A, C.
C. B, C, A.
D. C, A, B.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000061|!| ★

What is solution A called?

A. isotonic solution
B. hypotonic solution
C. hypertonic solution
D. none of the above

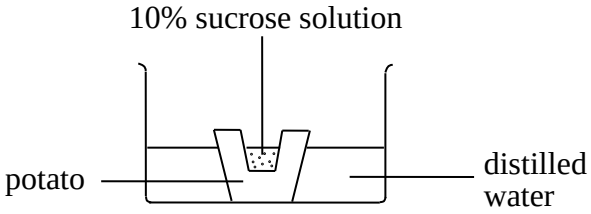
##A##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

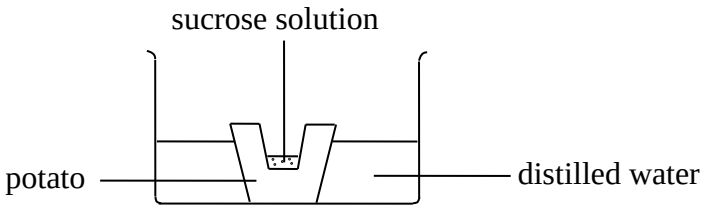
|!|BCEMB0040000062|!| ★ SI

Directions: The following three questions refer to the diagram below which shows an experimental set-up using a peeled potato which was cut into a cup shape.

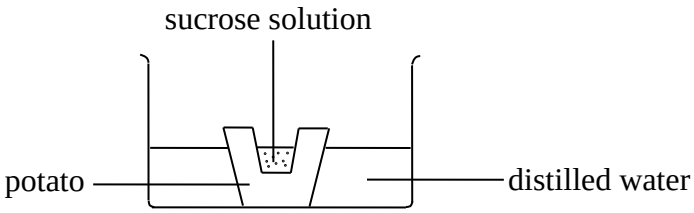


Which of the following set-ups shows the results of the experiment after four hours?

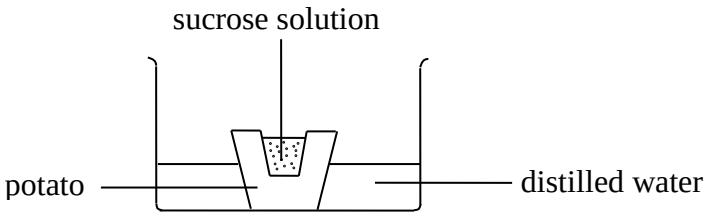
A.



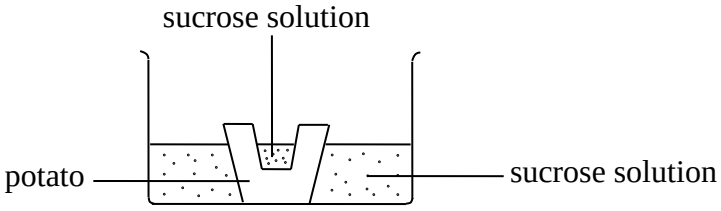
B.



C.



D.



□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000063|!| ★ **SI**

This experiment was performed to illustrate

A. the diffusion of solute molecules from a region of higher concentration to a region of lower concentration.
B. the net movement of water molecules from a region of higher water potential to a region of lower water potential across a differentially permeable membrane.
C. the transport of molecules against the concentration gradient.
D. the absorption of sucrose solution by potato tissues.

□

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0040000064|!| ★ **SI**

If a boiled potato cup is used instead of a fresh one, what will happen?

(1) Sucrose solution inside the cavity of the potato cup will diffuse out.
(2) Distilled water will move into the cavity of the potato cup.
(3) The liquid level inside the cavity of the potato cup will fall.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

□

##B##

Skill assessed	Mark
Applying concepts	1

© 2019 Aristo Educational Press Ltd.

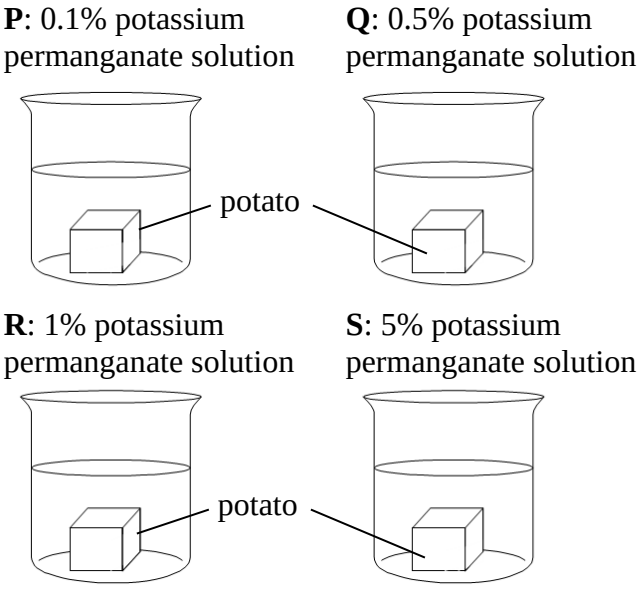
|!|BCEMA0040000065|!| ★★ **SI**

Directions: The following three questions refer to the two diagrams below. The diagrams illustrate two experiments demonstrating the movement of potassium permanganate into the potato tissues.

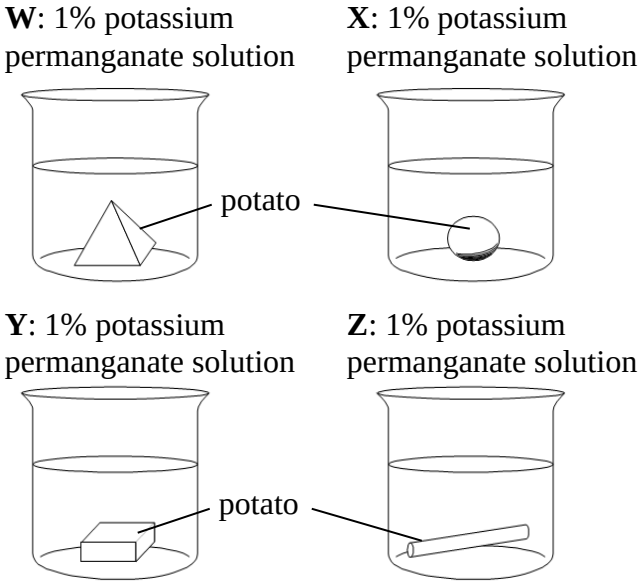
In experiment A, four identical potato cubes were placed separately in four beakers, each containing purple potassium permanganate solution of a different concentration.

In experiment B, four potato blocks with the same volume but different shapes were put into four beakers respectively. All of the beakers contained 1% potassium permanganate solution.

Experiment A



Experiment B



© 2019 Aristo Educational Press Ltd.

With reference to experiment A, if the potato cubes were cut after 20 minutes, the depth of penetration of the purple colour into the potato cube would be the greatest in

A. P.
B. Q.
C. R.
D. S.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0040000066|!| ★★ **SI**

Which of the following can be reflected by the results of experiment A?

A. This experiment demonstrates the process of active transport of potassium permanganate into cells.
B. The rate of osmosis increases with an increase in concentration of the potassium permanganate solution.
C. The rate of diffusion of the potassium permanganate solution increases with an increase in surface area of the potato cubes.
D. The rate of diffusion of potassium permanganate increases with an increase in concentration of the potassium permanganate solution.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0040000067|!| ★★ **SI**

Which of the following combinations about the factors that affect the rate of penetration of purple colour in the two experiments is correct?

	Experiment A	***Experiment B***
A.	mass of the potato cube	shape of the block
B.	temperature of the solution	surface area to volume ratio of the block
C.	surface area to volume ratio of the cube	concentration of the solution
D.	concentration of the solution	total surface area of the block

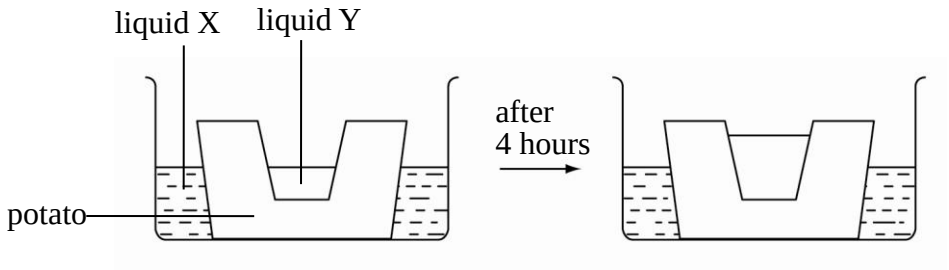
##D##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000068|!| ★ SI

Directions: The following three questions refer to the diagram below which shows an experiment using a peeled potato. The potato was cut into a cup shape.



This experiment is carried out to demonstrate

A. the absorption of dissolved substances by potato.
B. the movement of water molecules through living cells.
C. the diffusion of dissolved substances down the concentration gradient.
D. the transport of dissolved substances through cell membranes.

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000069|!| ★ SI

Which of the following combinations correctly identifies liquids X and Y?

	Liquid X	***Liquid Y***
A.	concentrated sucrose solution	dilute sucrose solution
B.	dilute sucrose solution	concentrated sucrose solution
C.	concentrated glucose solution	distilled water
D.	distilled water	dilute glucose solution

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000070|!| ★ SI

Which of the following shows the changes in the concentrations of liquids X and Y during the experiment?

	Liquid X	***Liquid Y***
A.	decreases	increases
B.	increases	decreases
C.	remains unchanged	decreases
D.	remains unchanged	remains unchanged

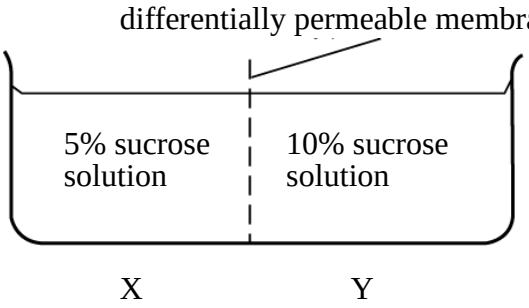
© 2019 Aristo Educational Press Ltd.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000071|!| ★

Directions: The following two questions refer to the diagram below, which shows an investigation using a differentially permeable membrane.



In the above set-up, water moves

A. across the membrane from X to Y only.
B. across the membrane from Y to X only.
C. in both directions but mainly from Y to X.
D. in both directions but mainly from X to Y.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000072|!| ★

The level of the solution in Y will

A. rise.
B. drop.
C. remain unchanged.
D. initially rise and then drop.

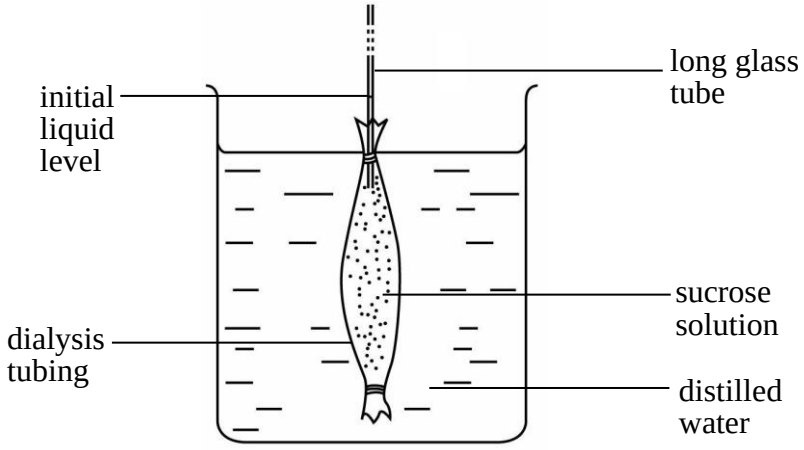
##A##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

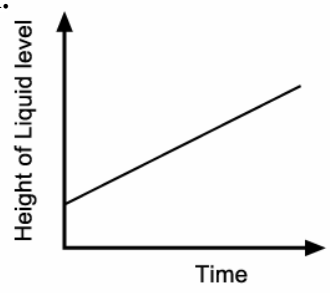
|!|BCEMB0040000073|!| ★ SI

Directions: The following three questions refer to the set-up below which is used to demonstrate osmosis.

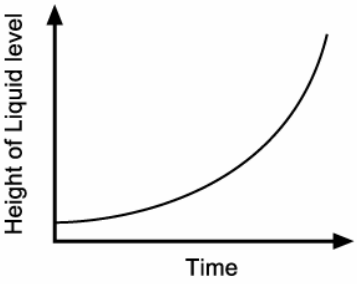


Which of the following graphs shows the change in the liquid level inside the glass tube?

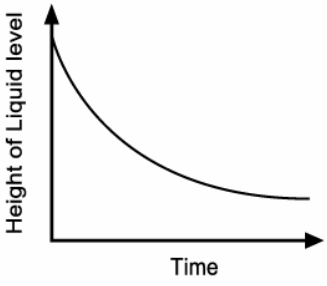
A.



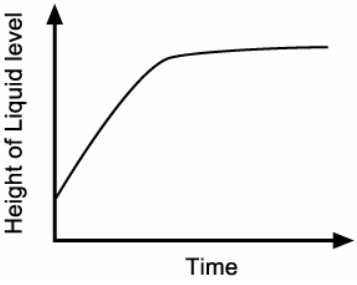
B.



C.



D.



□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000074|!| ★ SI

If glucose solution is used instead of sucrose solution, the liquid level will rise more slowly in the experiment. This is because

A. glucose is a reducing sugar but sucrose is not.
B. glucose molecules are small enough to pass through the tubing but sucrose molecules are not.
C. glucose is more soluble than sucrose in water.
D. glucose solution has a higher water potential than sucrose solution.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0040000075|!| ★ SI

If living tissue is used instead of the dialysis tubing to demonstrate osmosis, which of the following tissues should not be used?

A. pig bladder
B. chicken crop
C. fish swim bladder
D. mammalian skin

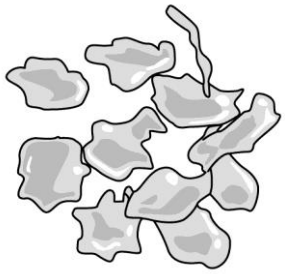
##D##

Skill assessed	Mark
Applying concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000076|!| ★

Directions: The following two questions refer to an experiment in which some human red blood cells were added into a liquid X. After a period of time, observation was made with a microscope and a diagram of the cells was drawn as follows.



Liquid X is

A. distilled water.
B. sea water.
C. physiological saline.
D. ice water.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000077|!| ★

Which of the following words can be used to describe the condition of plant cells if they are put into liquid X?

A. hydrolysis
B. turgidity
C. haemolysis
D. plasmolysis

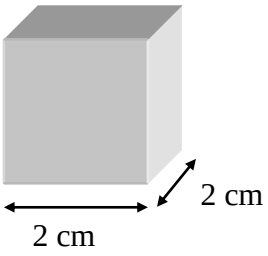
##D##

Skill assessed	Mark
Applying concepts	1

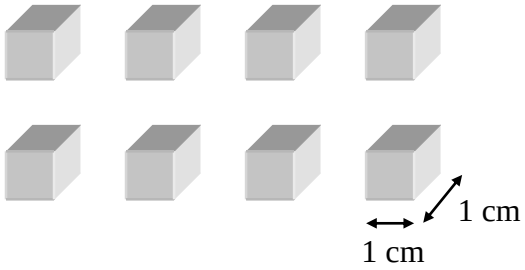
© 2019 Aristo Educational Press Ltd.

|!|BCEMA0040000078|!| ★★

Directions: The following two questions refer to an experiment below. In the experiment, a student studied the relationship between the surface area of cells and the rate of diffusion. He cut the potatoes into cubes of different sizes as shown below.



Cube A



Cubes B

Cube A and cubes B were put into a beaker of blue ink. After an hour, the depth of colour penetration in each cube was measured.

What are the surface area to volume ratios of cube A and cube B?

	Cube A	Cube B
A.	6:1	3:1
B.	3:1	6:1
C.	1:1	2:1
D.	2:1	1:1

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMA0040000079|!| ★★

Which of the following statements about the cubes is ***incorrect***?

A. The depth of colour penetration in cube B is less than that in cube A after one hour.

B. The volume of cube A and the total volumes of all cubes B are the same.

C. The rate of colour penetration in cube A is slower than that in cube B.

D. The total surface area of all cubes B is greater than that in cube A.

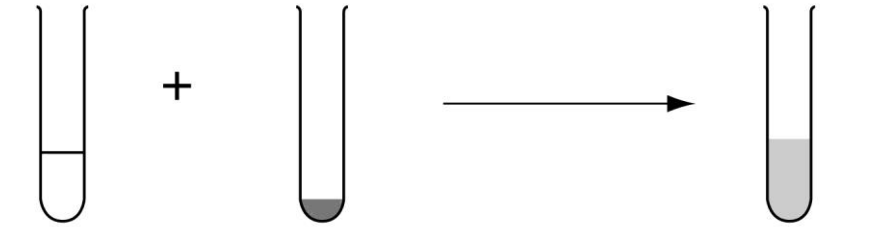
##A##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000080|!| ★ SI

Directions: The following two questions refer to the diagram below which shows an experiment. In the experiment, 5 cm^3 of liquid A was added to 1 cm^3 of fresh blood in a test tube. After 30 minutes, the mixture became clear red.



liquid A fresh blood clear red mixture

Which of the following is liquid A?

A. distilled water
B. isotonic glucose solution
C. 20% sucrose solution
D. sea water

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0040000081|!| ★ SI

Which of the following statements correctly explain(s) the results?

(1) The red colour is due to the presence of haemoglobin.
(2) Most of the cells in the mixture are not intact.
(3) Water in the red blood cells moves out to liquid A by osmosis.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

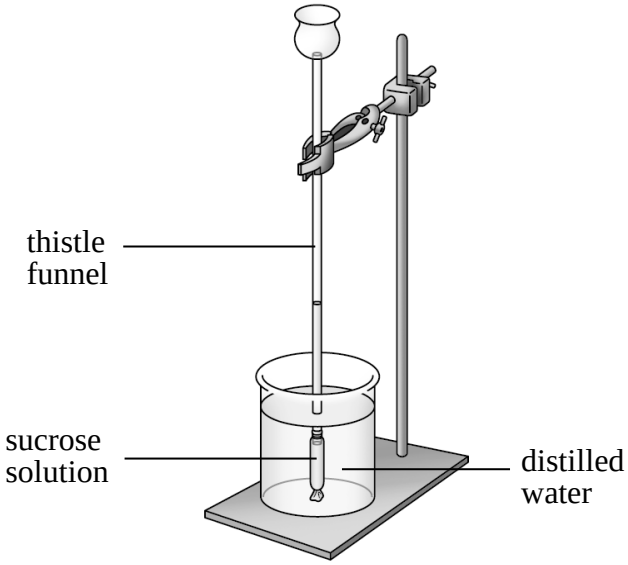
##C##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0040000082|!| **SI**

Directions: The following two questions refer to the diagram below. A dialysis tubing was tied to the end of the glass tube of a thistle funnel. The liquid inside the dialysis tubing was sucrose solution, while the liquid outside the tubing was distilled water. The liquid level in the thistle funnel rose to a certain height and then stopped.



The liquid level in the thistle funnel rose because of

A. simple diffusion.
B. osmosis.
C. active transport.
D. facilitated diffusion.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0040000083|!| **SI**

Which of the following statements correctly explains why the liquid level rose to a certain height and then stopped?

A. The rising force of the liquid level was balanced by the gravitational force.
B. The rate of water molecules leaving the dialysis tubing became equal to that of entering the tubing.
C. There were no water molecules entering the dialysis tubing anymore.
D. The atmospheric pressure suppressed the rising of the liquid level.

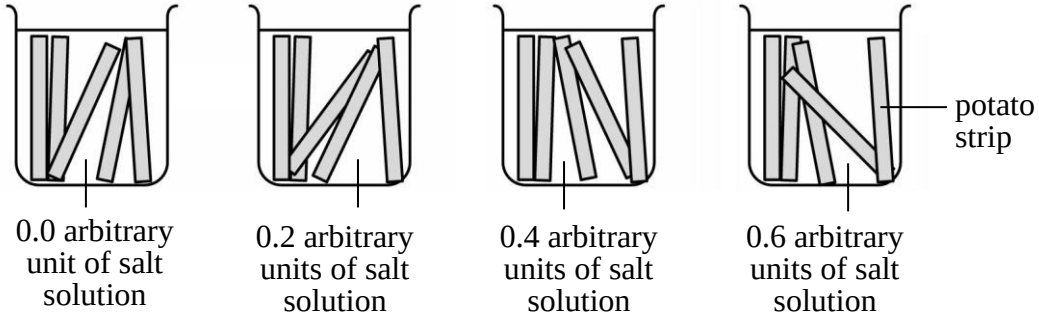
##B##

Skill assessed	Mark
Applying concepts	1

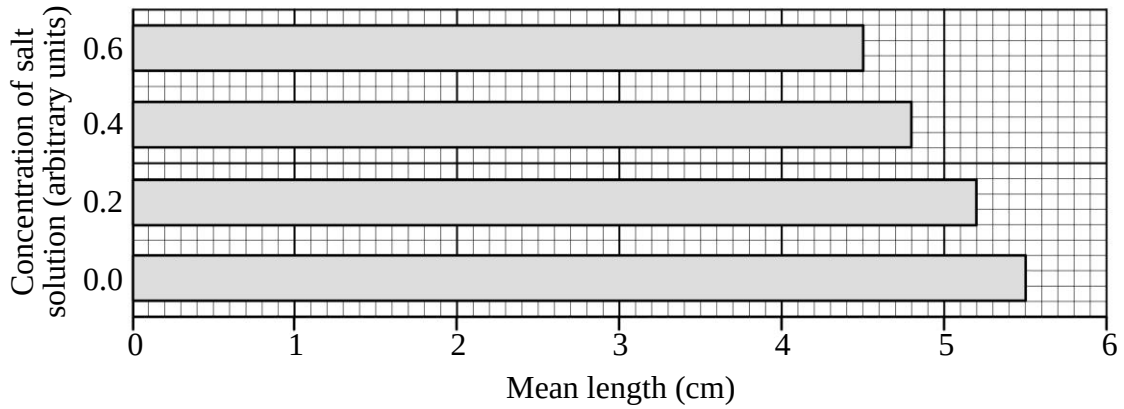
© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000084|!| ★

Directions: The following two questions refer to the experiment below. In the experiment, several thin strips with equal size were prepared from a peeled potato. All of them were 5 cm long. They were placed in salt solutions of different concentrations as shown below.



After two hours, the potato strips were taken out from the solutions. Their lengths were measured and the results are shown in the following chart.



Which process led to the change in the length of the potato strips?

A. diffusion
B. osmosis
C. active transport
D. diffusion and active transport

□

##B##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0040000085|!| ★

In which concentration of salt solution would you expect the length of the potato strip to remain unchanged?

A. 0.0 arbitrary units

B. 0.1 arbitrary units

C. 0.3 arbitrary units

D. 0.6 arbitrary units

##C##

Skill assessed	Mark
Analysing data	1